# Software Requirement Specification (SRS)

## StudySphere

AI-Powered Learning & Resource Sharing Platform

# 1. Introduction

## 1.1 Purpose

The purpose of this system is to build a comprehensive AI-powered Learning & Resource Sharing Platform that revolutionizes collaborative learning through intelligent features, gamification, and social engagement. The platform serves as an ecosystem where users can upload, access, and manage study resources while leveraging AI for personalized learning paths, practice modules, and real-time collaboration.

## 1.2 Scope

The platform provides the following core capabilities:

- Resource Sharing: Users upload resources (PDFs, blogs, links, notes) with admin verification, ratings, reviews, and advanced search capabilities.
- DSA Practice Module: Topic-wise practice with built-in code editor, multiple language support, automated test cases, video explanations, and progress tracking.
- AI Module: Generates personalized quizzes, conducts technical/non-technical interviews, provides AI study assistant, code review, and adaptive learning paths.
- Social & Collaboration: Study groups, discussion forums, peer-to-peer mentoring, real-time chat, and follow system.
- Gamification: Badges, achievements, leaderboards, streak tracking, points system, and weekly challenges.
- Analytics & Insights: Personal learning analytics, time tracking, weak area identification, and comprehensive progress visualization.
- User Profile: Comprehensive profile with personal details, uploaded resources, solved questions, quiz results, achievements, and activity history.
- Role-Based Access: Multi-tier role system (Admin, Moderator, User) with granular permissions.

## 1.3 Document Conventions

- Functional requirements are marked as FR-XXX
- Non-functional requirements are marked as NFR-XXX

- Priority levels: High, Medium, Low

# 2. Technologies & Architecture

## 2.1 Frontend Stack

- React 18+ with TypeScript for type safety
- React Router v6 for routing
- Zustand for state management
- TailwindCSS for styling
- shadcn/ui for component library
- Monaco Editor for code editing
- Socket.io Client for real-time features
- PWA capabilities for offline access

## 2.2 Backend Stack

- Node.js with Express.js framework
- MongoDB with Mongoose ODM
- Redis for caching and session management
- Socket.io for real-time communication
- Bull Queue for background job processing
- Docker for containerization

## 2.3 Authentication & Security

- Clerk Authentication for user management
- JWT tokens for API authentication
- Two-Factor Authentication (2FA) support
- OAuth integration (Google, GitHub)
- Rate limiting and API throttling

## 2.4 AI & ML Services

- OpenAI GPT-4 for content generation and AI assistant
- Custom ML models for learning path recommendations
- TensorFlow.js for client-side predictions
- Natural Language Processing for code review

## 2.5 Third-Party Integrations

- AWS S3 for file storage
- CloudFlare CDN for content delivery
- SendGrid for email notifications
- Firebase Cloud Messaging for push notifications

- Stripe for payment processing (premium features)
- Judge0 API for code execution

## 3. User Roles and Permissions

### 3.1 User (Standard Role)

**Access Rights:**

- Upload and manage personal resources
- View, rate, and review verified resources
- Bookmark and create resource collections
- Access DSA practice module with built-in editor
- Use AI features (quiz, interview, study assistant)
- Join/create study groups and participate in forums
- Follow other users and view their public profiles
- Earn badges, achievements, and track streaks
- Access personal analytics dashboard
- Participate in weekly challenges and competitions

### 3.2 Moderator (Elevated Role)

All User permissions plus:

- Review and verify pending resources
- Moderate discussion forums and comments
- Handle user reports and content flags
- Access basic moderation analytics

### 3.3 Admin (Full Access)

All Moderator permissions plus:

- Full platform management dashboard
- User management (view, block, unblock, delete)
- Assign/revoke Moderator and Admin roles
- Access comprehensive platform analytics
- Configure system settings and feature flags
- Manage DSA question database
- Export data and generate reports

# 4. Functional Requirements

## 4.1 Authentication & User Management (FR-001)

**Core Features:**

- Sign up/login via Clerk Authentication
- OAuth integration (Google, GitHub, LinkedIn)
- Two-Factor Authentication (2FA) support
- Email verification and password recovery
- Session management with automatic timeout
- Concurrent session detection and control

## 4.2 Resource Sharing Module (FR-002)

**Resource Management:**

- Upload resources with metadata (name, subject, type, tags, description)
- Support for PDFs, documents, links, videos, and text notes
- Version control for resource updates
- Admin/Moderator verification before publication
- Automated plagiarism detection
- OCR for scanned documents to make them searchable

**Discovery & Interaction:**

- Advanced search with filters (subject, type, rating, date, tags)
- Tag-based categorization and navigation
- Rating system (1-5 stars) with written reviews
- Bookmark resources for later access
- Create custom collections/playlists
- Comments and threaded discussions on resources
- Download statistics and popularity tracking
- Resource dependency mapping (prerequisites)

## 4.3 DSA Practice Module (FR-003)

**Problem Library:**

- Browse problems by topic (Arrays, Trees, Graphs, DP, etc.)
- Filter by difficulty (Easy, Medium, Hard)
- Track status (Solved, Attempted, Unsolved)
- Company-tagged problems
- AI-powered problem recommendations

**Coding Environment:**

- Built-in code editor (Monaco) with syntax highlighting
- Multi-language support (Python, Java, C++, JavaScript, etc.)

- Run code against test cases
- Submit solutions for automated evaluation
- Time and space complexity analysis
- AI-powered code review and suggestions
- Save multiple solution attempts

**Learning Resources:**

- Video explanations for each problem
- Editorial solutions with detailed walkthrough
- Similar problems suggestions
- Community solutions (peer review)
- Optional redirect to LeetCode for additional practice

## 4.4 AI-Powered Features (FR-004)

**AI Quiz Generator:**

- Generate quizzes based on subject, difficulty, and question count
- Multiple question types (MCQ, True/False, Fill in the blanks)
- Adaptive difficulty based on performance
- Timed quizzes with auto-submit
- Detailed score breakdown and explanations
- Quiz history and performance trends

**AI Interview Simulator:**

- Technical interviews (based on selected domain/language)
- Behavioral interviews (communication skills, problem-solving)
- Real-time feedback and scoring
- Detailed analysis report with improvement suggestions
- Voice-based or text-based interaction modes

**AI Study Assistant:**

- Chat-based help for specific topics
- Code debugging assistance
- Concept explanation with examples
- Step-by-step problem-solving guidance
- Context-aware responses based on user's learning history

**Personalized Learning Paths:**

- AI-generated study roadmaps based on goals
- Adaptive content recommendations
- Weak area identification and targeted practice
- Automated study schedule generation

## 4.5 Social & Collaboration Features (FR-005)

**Study Groups:**

- Create public/private study groups
- Group chat with real-time messaging
- Shared resource collections within groups
- Group challenges and competitions
- Member roles (Owner, Moderator, Member)

**Discussion Forums:**

- Topic-based discussion threads
- Upvote/downvote system
- Threaded replies with notifications
- Code snippets and LaTeX support
- Tag posts for easy discovery

**Peer Connections:**

- Follow/unfollow other users
- View follower/following lists
- Activity feed showing followed users' activities
- Direct messaging between users
- Peer-to-peer mentoring connections

## 4.6 Gamification System (FR-006)

**Achievement System:**

- Badges for milestones (First upload, 50 problems solved, 100-day streak)
- Achievement tiers (Bronze, Silver, Gold, Platinum)
- Special badges for unique accomplishments
- Achievement showcase on profile

**Points & Ranking:**

- Earn points for all platform activities
- Daily, weekly, and all-time leaderboards
- Category-specific leaderboards (DSA, Resources, Quizzes)
- Reputation system based on contribution quality
- Level progression system (Beginner → Expert → Master)

**Challenges & Competitions:**

- Weekly coding challenges
- Topic-specific contests
- Group competitions

- Prizes and rewards for top performers

**Streak Tracking:**

- Daily activity streaks
- Streak freeze power-ups
- Current and longest streak display
- Streak milestones and rewards

## 4.7 Analytics & Insights (FR-007)

**Personal Dashboard:**

- Overall progress visualization (charts and graphs)
- Time tracking across different subjects/topics
- Performance trends over time
- Strengths and weaknesses analysis
- Goal setting and tracking
- Personalized recommendations for improvement

**Progress Reports:**

- Weekly/monthly summary emails
- Export detailed progress reports as PDF
- Shareable achievement certificates
- Comparison with similar users (anonymized)

**Admin Analytics:**

- Platform-wide usage statistics
- User engagement metrics
- Resource popularity and quality metrics
- Moderation queue analytics
- Revenue and subscription metrics

## 4.8 Notification System (FR-008)

- Real-time in-app notifications
- Email notifications (configurable preferences)
- Push notifications (mobile/desktop)
- Notification categories: Resources, Mentions, Replies, Achievements, Challenges
- Weekly digest emails
- Study reminders and scheduled notifications
- Do Not Disturb mode

## 4.9 Enhanced User Profile (FR-009)

**Profile Information:**

- Personal details (name, bio, location)
- Social links (GitHub, LinkedIn, Twitter, portfolio)
- Profile picture and banner image
- Skills and interests tags
- Current learning goals

**Activity & Contributions:**

- Uploaded resources with edit/delete options
- DSA progress summary (problems solved by difficulty/topic)
- Quiz history and scores
- Badges and achievements display
- Activity heatmap (contribution calendar)
- Recent activity feed
- Statistics dashboard (total contributions, rank, level)

## 5. Non-Functional Requirements

### 5.1 Performance (NFR-001)

- Page load time < 2 seconds for 95% of requests
- API response time < 500ms for standard operations
- AI quiz generation < 5 seconds
- Code execution < 10 seconds
- Support for 10,000+ concurrent users

### 5.2 Scalability (NFR-002)

- Horizontal scaling for backend services
- Database sharding for large datasets
- CDN for static content delivery
- Load balancing across multiple servers
- Auto-scaling based on traffic patterns

### 5.3 Security (NFR-003)

- HTTPS/TLS encryption for all communications
- JWT token-based authentication with refresh tokens
- Rate limiting: 100 requests/minute per user
- OWASP Top 10 vulnerability protection
- Regular security audits and penetration testing

- Data encryption at rest and in transit
- GDPR and data privacy compliance
- Content Security Policy (CSP) headers

### 5.4 Reliability (NFR-004)

- 99.9% uptime SLA
- Automated daily database backups
- Point-in-time recovery capability
- Disaster recovery plan with RTO < 4 hours
- Comprehensive error logging and monitoring
- Automated health checks and alerting

### 5.5 Usability (NFR-005)

- Mobile-responsive design (iOS, Android, tablets)
- WCAG 2.1 Level AA accessibility compliance
- Dark mode and light mode support
- Progressive Web App (PWA) capabilities
- Offline mode for downloaded resources
- Multi-language support (English, Spanish, Hindi, etc.)
- Keyboard navigation and screen reader support

### 5.6 Maintainability (NFR-006)

- Comprehensive API documentation (Swagger/OpenAPI)
- Code coverage > 80% for critical modules
- Continuous Integration/Continuous Deployment (CI/CD)
- Modular architecture with clear separation of concerns
- Feature flags for gradual rollouts
- Detailed deployment and rollback procedures

## 6. Database Schema & Models

### 6.1 User Model

- userId (from Clerk, primary key)
- name, email, profilePicture, bannerImage
- bio, location, skills (array), interests (array)
- role (User/Moderator/Admin)
- socialLinks (GitHub, LinkedIn, Twitter, portfolio)

- isBlocked, blockReason, blockedAt
- uploadedResources (array of resource IDs)
- bookmarkedResources (array of resource IDs)
- collections (array of collection objects)
- quizHistory (array of quiz result objects)
- dsaProgress (nested object with topics/difficulties)
- achievements (array of badge objects)
- points, level, rank
- currentStreak, longestStreak, lastActiveDate
- followers (array of user IDs)
- following (array of user IDs)
- studyGroups (array of group IDs)
- notificationPreferences (object)
- createdAt, updatedAt

## 6.2 Resource Model

- resourceId (primary key)
- resourceName, description
- subject, tags (array)
- type (pdf/doc/link/video/text)
- fileUrl, thumbnailUrl, fileSize
- uploadedBy (user ID)
- status (Pending/Approved/Rejected)
- verifiedBy (moderator/admin user ID), verifiedAt
- ratings (array of rating objects: userId, rating, review, timestamp)
- averageRating, totalRatings
- views, downloads, bookmarks (counts)
- version, previousVersions (array)
- prerequisites (array of resource IDs)
- comments (array of comment objects)
- createdAt, updatedAt

## 6.3 DSA Question Model

- questionId (primary key)
- title, description, exampleInput, exampleOutput
- topic (Arrays/Strings/Trees/Graphs/DP/etc.)
- difficulty (Easy/Medium/Hard)
- companies (array of company names)
- tags (array)
- constraints, hints (array)
- testCases (array of input/output pairs)
- starterCode (object with language keys)
- editorial (solution explanation with code)
- videoUrl (explanation video)

- similarQuestions (array of question IDs)
- leetcodeUrl (optional redirect)
- createdAt, updatedAt

## 6.4 Submission Model

- submissionId (primary key)
- userId, questionId
- code, language
- status (Accepted/Wrong Answer/TLE/Runtime Error)
- testCasesPassed, totalTestCases
- runtime, memory
- aiReview (feedback from AI code review)
- submittedAt

## 6.5 Quiz Model

- quizId (primary key)
- subject, difficulty
- questions (array of question objects)
- userId
- score, totalQuestions, correctAnswers
- timeLimit, timeTaken
- userAnswers (array)
- completedAt, createdAt

## 6.6 Study Group Model

- groupId (primary key)
- name, description, avatar
- isPrivate (boolean)
- ownerId (user ID)
- moderators (array of user IDs)
- members (array of user IDs)
- sharedResources (array of resource IDs)
- tags (array)
- createdAt, updatedAt

## 6.7 Discussion/Forum Model

- postId (primary key)
- title, content
- authorId (user ID)
- category, tags (array)
- upvotes, downvotes (counts)
- votedBy (array of objects: userId, voteType)

- replies (array of reply objects)
- isPinned, isLocked
- createdAt, updatedAt

### 6.8 Notification Model

- notificationId (primary key)
- recipientId (user ID)
- type (Resource/Mention/Reply/Achievement/Challenge/System)
- title, message
- relatedId (ID of related entity)
- isRead, readAt
- createdAt

## 7. System Architecture & Flow

### 7.1 User Journey Flow

1. User registers/logs in → Clerk Auth verifies identity
2. User profile created/synced in MongoDB database
3. User accesses modules: Resource sharing, DSA practice, AI features, Social
4. Admin/Moderators verify pending resources
5. AI generates personalized content (quizzes, interviews, recommendations)
6. User interactions tracked → Analytics updated → Progress reflected
7. Achievements unlocked → Notifications sent → Gamification rewards

### 7.2 Caching Strategy

- Redis cache for frequently accessed data (user profiles, popular resources)
- Cache invalidation on data updates
- Session storage in Redis
- CDN caching for static resources (images, videos, PDFs)

### 7.3 Background Job Processing

- Email notifications (queued via Bull)
- Analytics aggregation (daily/weekly)
- AI content generation (quizzes, recommendations)
- File processing (OCR, thumbnail generation)
- Database cleanup and maintenance tasks

# 8. Key API Endpoints (Sample)

### Authentication

- POST /api/auth/register - User registration
- POST /api/auth/login - User login
- POST /api/auth/logout - User logout

### Resources

- GET /api/resources - Get all verified resources (with filters)
- GET /api/resources/:id - Get resource details
- POST /api/resources - Upload new resource
- PUT /api/resources/:id - Update resource
- DELETE /api/resources/:id - Delete resource
- POST /api/resources/:id/rate - Rate a resource

### DSA

- GET /api/dsa/questions - Get all questions (with filters)
- GET /api/dsa/questions/:id - Get question details
- POST /api/dsa/submit - Submit solution
- GET /api/dsa/progress - Get user's DSA progress

### AI Features

- POST /api/ai/quiz/generate - Generate quiz
- POST /api/ai/interview/start - Start interview session
- POST /api/ai/assistant/chat - Chat with AI study assistant
- POST /api/ai/recommend - Get personalized recommendations

### Social

- GET /api/groups - Get all study groups
- POST /api/groups - Create study group
- GET /api/forums - Get forum posts
- POST /api/users/:id/follow - Follow a user

### Admin

- GET /api/admin/dashboard - Admin dashboard statistics
- GET /api/admin/users - Get all users
- PUT /api/admin/resources/:id/verify - Verify resource
- PUT /api/admin/users/:id/block - Block user

# 9. Future Enhancements & Roadmap

## 9.1 Short-term Enhancements (6-12 months)

- Native mobile applications (iOS & Android)
- Video conferencing for study groups
- Live coding sessions with screen sharing
- Integration with LMS platforms (Canvas, Moodle)
- API marketplace for third-party integrations
- Premium subscription tier with advanced features

### 9.2 Long-term Vision (12-24 months)

- AI-powered virtual tutor with voice interaction
- Blockchain-based certification system
- Integration with job boards for placement opportunities
- VR/AR learning experiences for complex concepts
- Multi-tenancy support for educational institutions

## 10. Appendices

### 10.1 Glossary

- DSA - Data Structures and Algorithms
- SRS - Software Requirements Specification
- PWA - Progressive Web Application
- CDN - Content Delivery Network
- JWT - JSON Web Token
- WCAG - Web Content Accessibility Guidelines

### 10.2 References

- Clerk Authentication Documentation
- MongoDB Best Practices
- OpenAI API Documentation
- React and Next.js Performance Guidelines
- OWASP Security Best Practices

### 10.3 Document Information

- Document Type: Software Requirements Specification
- Platform Name: StudySphere
- Last Updated: February 2026
- Status: Active Development

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*End of Document*